Subsea Integrity Copilot - Customer Brief

Customer, problem and users. NorthSea Integrity Operations is a fictional offshore oil-and-gas operator; this demo focuses on its gas pipelines. Monitoring, integrity and instrumentation engineers must distinguish a possible leak from sensor faults or shared telemetry outages, then choose what to inspect. The agent is a read-only evidence layer over existing monitoring workflows, not a replacement dashboard or safety system.

Why I chose this problem. It builds on my PhD research in marine sensor data quality and anomaly detection. The practical question is whether unusual readings point to a real pipeline problem or faulty sensors. This is a useful FDE case because engineers need evidence they can check before deciding what to inspect, not just another alarm or chatbot answer. Leak localization is a simplified demo extension, not a capability validated by my thesis.

Why the prospect has not signed. Their existing dashboard is cheaper, and errors have two costly sides. False alarms can trigger unnecessary offshore personnel, vessel or ROV mobilization and operational disruption. Missed releases risk environmental harm; oil-line spills also bring cleanup and liability. Serious regulatory violations can put licences at risk, including potential revocation under applicable law. They need evidence that AI reduces investigation effort without increasing missed-event risk, integration burden or data exposure.

Business stakeholder: Head of Subsea Operations, budget owner. Cares about inspection cost, delivery time, adoption, environmental exposure and accountability. Main objections: "Will fewer alarms conceal real leaks?", "What value can four weeks demonstrate?" and "Who owns a wrong recommendation?" The ask is a bounded shadow pilot alongside existing safety procedures, not autonomous intervention.

Technical stakeholder: Lead Subsea Integrity & Instrumentation Engineer. Cares about calibration, provenance, stale/incomplete/inconsistent data, environmental confounders and reproducible physical assumptions. Main objections: "Could these sensors share a failed gateway?", "Could currents or acoustic noise explain this?" and "Can I reproduce the location from raw evidence?"

What the demo demonstrates. An original synthetic network of 24 sensor nodes streams acoustic relative power, dissolved methane and temperature through a 60-second window advancing every five simulated seconds. Data includes gaps, spikes, mixed units/time formats and conflicting duplicates. Three scoped agents answer: (1) possible leak or instrumentation issue? (2) which pipeline section should an engineer inspect first, and why? (3) which sensors are unreliable and what evidence is missing? Numerical tools own normalization, corroboration and fitting; agents interpret checked evidence. Raw records, pinned answers and execution receipts are inspectable. A shared outage demonstrates abstention, not an all-clear.

Assumptions and limitations. The customer, network and telemetry are synthetic; no confidential employer data is used. Methane/acoustic signatures model gas, not validated oil-spill detection. Geometry, baselines and failure domains are assumed. Acoustic distance-decay fitting provides a compatible inspection candidate, not an exact leak position or calibrated confidence interval. Synthetic qualification is not field reliability. Missingness does not prove a leak or a particular hardware failure. Copilot inference is hosted, not local-only; only synthetic evidence is sent. No live historian/platform connector, certified detection, shutdown/control or production security certification is claimed. Human engineers retain inspection and safety authority.

Proposed next step. A conditional four-week shadow pilot on one network after data/security approval. Compare triage effort and reviewed false alarms against existing workflows; separately assess missed-event sensitivity with labeled events or validated tests. Track coverage and latency; poor coverage is not a healthy pipeline. Agree pricing, savings assumptions and safety/acceptance gates before work.

Private demo repository: https://github.com/ntnguyen-so/subsea-integrity-copilot